

Rollno :241001307

SOLVE A GRAPH USING DEPTH FIRST SEARCH(DFS)

Python code:

```
def dfs(graph, start, visited=None):  
    if visited is None:  
        visited = set()  
  
    visited.add(start)  
    print(start, end=" ")  
  
    for neighbor in graph[start]:  
        if neighbor not in visited:  
            dfs(graph, neighbor, visited)  
  
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],
```

```
'D': [],  
'E': ['F'],  
'F': []  
}
```

```
print("DFS traversal from node A:")  
dfs(graph, 'A')
```

output:

DFS traversal from node A:

A B D E F C

Result:

Thus the program of implementing DFS traversal on a graph to explore all nodes was executed successfully.